

Comprehensive Evaluation of RAG Knowledge Bases for Smoking Cessation Counseling

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Abstract

We present a rigorous comparison of Retrieval-Augmented Generation (RAG) approaches for smoking cessation counseling, using a novel independent test set validated for zero data leakage. Our analysis includes multiple metrics (ROUGE-L, BLEU, METEOR) across five independent trials with statistical significance testing.

Key Finding: Contrary to expectations, the baseline approach (no retrieval) outperformed all RAG variants (ROUGE-L: 0.231 vs 0.224 for best RAG). Among RAG approaches, AI-generated knowledge bases slightly exceeded human-curated ones (102.1% relative performance), meeting the first research target. However, the improvement over raw sources (+7.8%) fell short of the 40% target.

Methodological Contribution: We discovered severe data leakage in the original test set (7/15 questions with $\geq 90\%$ similarity to training data), invalidating previous results. This highlights the critical importance of test set independence verification in NLP research.

1 Methodology

1.1 Data Leakage Discovery

During our audit, we discovered that the original test set had significant overlap with training data:

Table 1: Original Test Set Data Leakage Analysis

Severity Level	Count	Percentage
Severe ($\geq 90\%$ similarity)	7	47%
Moderate (70–90% similarity)	6	40%
Clean ($< 70\%$ similarity)	2	13%

Four test questions were *exact matches* (100% similarity) to questions in the training data, fundamentally invalidating the evaluation.

1.2 Independent Test Set Creation

We created a new 15-question test set with:

- Maximum 50% similarity to any training question (fuzzy string matching)
- Coverage of all original topics (benefits, cravings, NRT, relapse, etc.)
- Grammatically distinct formulations
- Domain expert-validated reference answers

1.3 Evaluation Protocol

- **Model:** GPT-4o-mini (temperature=0.3, max_tokens=200)
- **Retrieval:** ChromaDB with cosine similarity, k=3 documents
- **Runs:** 5 independent trials
- **Statistical Analysis:** 95% confidence intervals, paired t-tests, Cohen's d

2 Results

2.1 Main Results

Table 2: RAG Approach Comparison (5 runs, 15 questions each)

Approach	ROUGE-L	BLEU	METEOR
Baseline (No RAG)	0.231 $\pm$ 0.008	0.054 $\pm$ 0.004	0.310 $\pm$ 0.010
Raw Sources RAG	0.208 $\pm$ 0.004	0.039 $\pm$ 0.003	0.275 $\pm$ 0.008
AI-Generated RAG	0.224 $\pm$ 0.005	0.057 $\pm$ 0.003	0.290 $\pm$ 0.008
Human-Curated RAG	0.220 $\pm$ 0.004	0.045 $\pm$ 0.003	0.296 $\pm$ 0.010

Values shown as mean $\pm$ 95% CI.

2.2 Statistical Comparisons (ROUGE-L)

* $p < 0.05$, ** $p < 0.01$

Table 3: Pairwise Statistical Comparisons

Comparison	Difference	Cohen’s d	p-value	Significant
AI vs Human	+2.1%	0.72	0.184	No
AI vs Raw	+7.8%	3.19	0.002	Yes**
Human vs Raw	+5.5%	1.89	0.014	Yes*

Table 4: Research Target Evaluation

Target	Expected	Actual	Status
AI achieves 84% of Human	$\geq 84\%$	102.1%	MET
AI 40% better than Raw	$\geq +40\%$	+7.8%	NOT MET

2.3 Professor’s Targets Assessment

3 Discussion

3.1 Key Findings

1. Baseline outperforms RAG: The most surprising result is that the no-retrieval baseline achieved the highest ROUGE-L score (0.231), suggesting that for well-documented domains like smoking cessation, the LLM’s parametric knowledge may be sufficient.

2. AI-generated matches human-curated: AI-generated knowledge bases performed comparably to human-curated ones (102.1% relative performance), with no statistically significant difference ($p = 0.184$).

3. Processed data beats raw sources: Both AI-generated and human-curated RAG significantly outperformed raw sources ($p < 0.05$), confirming the value of structured knowledge bases.

4. Data leakage reversed conclusions: The original test set showed Raw > Human > AI, which completely reversed with proper independence validation.

3.2 Why This Matters

The discovery of data leakage in the original test set demonstrates:

1. The importance of rigorous test set validation in NLP research
2. That high performance scores may be artifacts of data contamination
3. The need for fuzzy string matching during test set creation

3.3 Implications for RAG Research

Our findings suggest a domain-dependency hypothesis: RAG benefit may be *inversely related* to domain representation in LLM training data. For well-

documented topics:

- Baseline LLM knowledge may be comprehensive
- Retrieved context may introduce noise rather than signal
- RAG overhead may not justify marginal improvements

Future work should test this hypothesis across domains with varying LLM knowledge saturation.

3.4 Limitations

- Single domain (smoking cessation) - results may not generalize
- Single LLM (GPT-4o-mini) - other models may show different patterns
- 15 test questions - larger test sets would provide more statistical power
- Domain is well-covered in LLM training data - may reduce RAG benefit
- Metrics measure lexical overlap, not semantic accuracy or factual correctness

4 Conclusion

This study makes two primary contributions:

Methodological: We demonstrate the critical importance of test set independence verification. Data leakage completely reversed our conclusions, serving as a warning for NLP researchers.

Empirical: We find that RAG does not universally improve performance. For well-known domains, LLM parametric knowledge may suffice. When RAG is used, AI-generated knowledge bases perform comparably to human-curated ones at a fraction of the cost.

These findings shift the research question from “Does RAG help?” to “When does RAG help?” - a more nuanced and practically valuable inquiry.

Acknowledgments

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A Effect Size Reference

Cohen’s d interpretation:

- $|d| < 0.2$: Negligible

- $0.2 \leq |d| < 0.5$: Small
- $0.5 \leq |d| < 0.8$: Medium
- $|d| \geq 0.8$: Large

B Complete Metrics by Approach

Table 5: Complete Metrics (Mean $\pm$ 95% CI)

Approach	ROUGE-1	ROUGE-2	ROUGE-L	BLEU	METEOR
Baseline	0.342 ± 0.010	0.097 ± 0.012	0.231 ± 0.008	0.054 ± 0.004	0.310 ± 0.010
Raw	0.316 ± 0.005	0.068 ± 0.004	0.208 ± 0.004	0.039 ± 0.003	0.275 ± 0.008
AI	0.325 ± 0.004	0.086 ± 0.004	0.224 ± 0.005	0.057 ± 0.003	0.290 ± 0.008
Human	0.334 ± 0.007	0.079 ± 0.005	0.220 ± 0.004	0.045 ± 0.003	0.296 ± 0.010